UNIT 1 · TOPIC 1.4

AI-Based Cybersecurity Attacks

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. In a scenario where a relative receives a phone call from someone who sounds exactly like their family member asking for emergency bail money, yet the family member never made that call, which type of AI-powered attack does this illustrate?

- ○ A. AI voice cloning used to impersonate the victim
- ○ B. Automated vulnerability scanning
- ○ C. AI-generated phishing email
- ○ D. Deepfake video manipulation

2. Which of the following best describes AI-generated phishing?

- ○ A. A human attacker manually writing each phishing email one by one
- ○ B. The use of AI tools to automatically produce large volumes of personalized, convincing phishing messages
- ○ C. Software that blocks suspicious emails before they reach the inbox
- ○ D. A hardware device that intercepts network traffic between a sender and recipient

3. An adversary creates a realistic video of a company executive announcing a fake wire transfer authorization and sends it to the finance team. What category of AI-based attack does this represent?

- ○ A. Brute-force attack
- ○ B. SQL injection
- ○ C. Zero-day exploit
- ○ D. Deepfake

4. An employee receives an urgent email that uses their name, references a current internal project, and mimics their manager's writing style, asking them to click a link to approve a vendor payment. Which capability of AI tools most directly enables this level of personalization at scale?

- ○ A. AI tools encrypt phishing links to evade spam filters
- ○ B. Machine learning models brute-force email passwords to extract internal details before sending phishing messages
- ○ C. AI tools route phishing emails through multiple proxy servers to obscure their origin
- ○ D. Large language models generate contextually coherent, targeted text by processing scraped organizational data such as employee profiles, project names, and public communications

5. Explain how AI-powered voice cloning attacks generate risk for both individuals and organizations.

- ○ A. Voice cloning attacks generate risk solely by overwhelming phone networks with spam calls, degrading call quality
- ○ B. Voice cloning generates risk only when combined with simultaneous physical theft of the impersonated person's device
- ○ C. Voice cloning poses risk only if the victim's voicemail PIN has already been compromised

○ D. By enabling adversaries to impersonate trusted individuals in real-time phone calls, voice cloning can trigger unauthorized financial transfers, false emergency responses, or disclosure of sensitive information by deceived victims

6. How do automated AI attack tools change the threat landscape compared to traditional manual cyberattacks?

○ A. Automated AI tools eliminate the need for adversaries to have any technical knowledge, making cyberattacks equally accessible to all

○ B. AI tools allow adversaries to conduct more attacks simultaneously, target more victims, and adapt attack content faster than manual approaches permit

○ C. AI tools reduce attack scale because they depend on limited training datasets and cannot generalize to new targets

○ D. Automated attack tools are effective only against large organizations since small targets lack sufficient data to fuel AI models

7. A security analyst observes a phishing campaign in which messages that were clicked in one wave are used to retrain the generation model for the next wave. Which combination of risks does this adaptive feedback loop introduce that a static phishing campaign does not?

○ A. It allows the adversary to evolve attack content in near-real-time, systematically evading signature-based filters and improving click rates as the model learns which content bypasses current defenses

○ B. It increases bandwidth consumption and degrades corporate email infrastructure performance

○ C. It exposes the adversary to legal liability since retraining on click-interaction data constitutes unlawful processing of personal information in all jurisdictions

○ D. It limits the attack to organizations that rely primarily on email, excluding adversaries targeting other communication channels

8. In the voice-cloning scenario, the adversary accepted a social media connection request from the target and harvested voice samples over two weeks before executing the attack. Evaluate which combination of technical and behavioral vulnerabilities the adversary exploited across the full attack chain.

○ A. Technical: unencrypted phone network protocols; Behavioral: victim posted videos without audio watermarking

○ B. Technical: the cloning tool exploited a zero-day vulnerability in the social media platform's audio API; Behavioral: victim posted long-form podcasts that provided higher-fidelity voice samples

○ C. Technical: weak password on the victim's social media account allowing the adversary to access private posts; Behavioral: the relative failed to recognize the caller's phone number as unknown

○ D. Technical: absence of AI-voice-detection capability on the recipient's phone; Behavioral: victim accepted a connection request from an unverified contact and posted voice-sampable public video; recipient did not verify the caller's identity through an independent channel before complying

9. An AI tool generates phishing emails by scraping a target organization's public GitHub repositories, employee LinkedIn profiles, and press releases to model internal project vocabulary and reporting relationships. Evaluate how this data-aggregation technique changes the risk profile of the resulting attack relative to a generic phishing campaign.

○ A. The risk profile is essentially unchanged; adequately trained users can still distinguish AI-generated from human-written emails

○ B. Risk is lower because AI-generated content lacks the emotional authenticity of human writing and is therefore easier to flag as suspicious

○ C. Risk only increases for organizations with more than 500 employees, since smaller organizations produce insufficient public data to meaningfully train a model

○ D. The aggregated-data approach significantly raises the risk profile: contextual accuracy, correct project names, authentic relationship references, plausible internal vocabulary, defeats the user's most reliable recognition heuristic and substantially increases compliance probability before technical controls can intercept

10. Which of the following is a security control that can help protect individuals from AI-powered voice cloning scams like the one in which a relative was deceived into wiring money?

○ A. Establishing a family-specific code word known only to trusted contacts, used to verify the identity of any caller making an urgent request

○ B. Disabling automatic software updates on all devices

○ C. Encrypting the hard drive of the intended impersonation victim's computer

○ D. Enabling location services on the victim's mobile device

11. To reduce the risk of an adversary harvesting voice samples from public social media to fuel a cloning attack, which control is most directly relevant?

○ A. Reviewing and restricting the audience of video posts that contain the user's voice

○ B. Installing antivirus software on all devices used to access social media

○ C. Using a strong, unique password on the social media account

○ D. Enabling two-factor authentication on linked email accounts

12. Which security control is specifically designed to help verify whether an email was actually sent by the domain it claims to be from, a check directly relevant to AI-generated phishing that spoofs legitimate senders?

○ A. Full-disk encryption on the recipient's computer

○ B. Email authentication protocols such as SPF, DKIM, and DMARC

○ C. A VPN connection when checking email

○ D. Endpoint antivirus scanning of email attachments

13. Explain how AI-powered email filtering tools mitigate the risk of AI-generated phishing at a higher level of effectiveness than traditional signature-based spam filters.

○ A. AI filters detect AI-generated emails by analyzing keyboard timing metadata embedded in email headers

○ B. AI-powered filters analyze behavioral and semantic patterns, unusual sending cadences, linguistic drift from the claimed sender's history, anomalous link structures, enabling detection of AI-generated phishing even when the content lacks any previously catalogued malicious signature

○ C. AI filters rely on the same blacklisted-IP mechanism as traditional filters, but execute the lookup faster

○ D. AI filters prevent phishing by encrypting all inbound email automatically, making embedded links unclickable

14. A company wants to protect employees from deepfake video calls in which an adversary impersonates an executive to authorize fraudulent financial transactions. Which combination of security controls best addresses this specific threat?

○ A. Firewall rules that block video conferencing ports, combined with mandatory antivirus scans on all incoming video files

○ B. Multi-person authorization policies requiring more than one approver for high-value transactions, combined with out-of-band identity verification before acting on instructions received via video call

○ C. Requiring all executives to use personal email accounts for sensitive financial communications instead of corporate video platforms

○ D. Installing deepfake-detection software on employee devices and blocking all video conferencing platforms

15. Determine which security control primarily addresses the social engineering vulnerability exploited in AI-augmented attacks, as distinct from technical or infrastructure vulnerabilities.

- ○ A. Security awareness training that specifically covers AI-powered attack techniques, voice cloning, deepfakes, AI-generated phishing, and trains users to apply independent verification before acting on urgent requests
- ○ B. Patching the operating system and all installed applications to the latest available version
- ○ C. Enabling full-disk encryption on all company-issued devices
- ○ D. Deploying a web application firewall in front of all public-facing services

16. A security architect proposes the following control set to defend financial authorization workflows against AI voice cloning: (1) Deploy voice-biometric authentication on all inbound calls to the finance department. (2) Train employees to question urgent requests made by phone. (3) Log all inbound calls to the finance department. Evaluate this control set and identify the most critical gap it leaves unaddressed.

- ○ A. The set omits full-disk encryption, which is required for call log integrity
- ○ B. Logging inbound calls violates data protection regulations in most jurisdictions, making Control 3 legally impermissible
- ○ C. Voice-biometric authentication is itself potentially vulnerable to a sufficiently high-fidelity voice clone; if the biometric model has not been adversarially evaluated against AI-generated voice samples, Control 1 may provide false assurance rather than genuine verification
- ○ D. The three controls are redundant with one another and can be simplified without any reduction in coverage

17. An organization implements the following layered control set against AI-generated phishing: (1) DMARC, DKIM, and SPF email authentication. (2) AI-powered behavioral email filtering. (3) Mandatory security awareness training updated once per year. (4) Multi-factor authentication on all accounts. Evaluate which residual risk remains most significant despite this control set.

- ○ A. A highly contextualized, low-volume spear-phishing message sent from a legitimately authenticated domain, or a lookalike domain just outside SPF/DKIM coverage, can satisfy both authentication and behavioral anomaly thresholds; annual training may not have equipped users to detect contextually accurate, AI-generated content that the technical layers have already passed
- ○ B. DMARC, DKIM, and SPF operate only on inbound email and provide no protection against reply-chain hijacking attacks
- ○ C. MFA cannot be deployed on corporate email accounts, making it ineffective against phishing regardless of how it is configured
- ○ D. AI-powered behavioral filters require a minimum of six months of baseline traffic data, during which the organization has no phishing protection at all

18. A security team is evaluating whether to invest in a real-time AI deepfake-detection tool for their video conferencing platform versus strengthening procedural controls, multi-person authorization, callback verification, and out-of-band approval workflows. Evaluate the relative long-term durability of these two approaches given the co-evolutionary nature of AI generation and detection.

- ○ A. AI deepfake detection and generation models co-evolve in an adversarial arms race; detection capability at any point in time may be outpaced by advances in generation. Procedural controls impose a verification burden that is independent of deepfake quality and therefore provide more durable protection, though they introduce friction and may not scale to all interaction types
- ○ B. Procedural controls are unambiguously superior because AI detection tools have been proven ineffective in all real-world deepfake deployment scenarios to date
- ○ C. The AI detection tool is unambiguously superior because it operates at machine speed and does not depend on human judgment, which is inherently error-prone

○ D. Neither approach is effective against sophisticated adversaries; organizations should ban video conferencing and replace it with in-person meetings for all sensitive decisions